Jordi Laguarta Soler

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I build multimodal ML systems end to end for domains where expert labels are scarce: foundation models as annotators, self-supervised pretraining, explainable predictions, and national-scale inference evaluated under distribution shift. At MIT and Microsoft Research, I have applied them to high-stakes problems in health and food security: screening for COVID-19 and Alzheimer's from coughs and speech, and mapping crops across entire countries. I lead research on evaluation science for geospatial foundation models. I have mentored student researchers for over three years. PhD expected February 2027; seeking research and applied scientist roles, open to forward-deployed and research-product work.

EXPERIENCE

MIT Earth Intelligence Lab · PhD researcher (advisor: Prof. Sherrie Wang) Sep 2022 – present

- **National crop-type map of India** (first author, manuscript in preparation). Built a VLM data engine: Gemini 3.1 Pro generated 728,490 crop labels from street-level images across 737 districts (90.2% agreement with agronomists, six classes). A classifier on AlphaEarth satellite embeddings maps every 10 m pixel of India at 77.1% overall accuracy on expert-labelled test points; district crop areas track official statistics (r = 0.60–0.96).
- Chose the embedding classifier on an accuracy–inference-cost trade-off: national inference in under 4 hours on Earth Engine, versus an estimated 1–2 years for a time-series CNN that reaches 85.1% on the same test set.
- **Thailand crop-type map** (first author, AAAI 2024). GPT-4V zero-shot labels on Google Street View images replaced field surveys and expert annotation as the training signal for a national 10 m map of four crops: 92% accuracy, within one point of the expert-supervised map (93%).
- **Benchmark reliability for geospatial foundation models.** Lead research on whether Earth-observation benchmarks resolve real differences between foundation models beyond seed variance, comparing Panopticon, AnySat, DINOv3, DOFA and ScaleMAE across tasks and training-set sizes.
- Mentored three student researchers.

Microsoft Research · Research intern, Seattle May 2024 – Aug 2024

- Designed and pretrained a self-supervised multimodal satellite foundation model for long time series (Sentinel-1, Sentinel-2, Dynamic World) for the AI for Agriculture and AI for Industry teams. Processed 20 TB of imagery, optimised data-streaming throughput and parallelised training across multiple GPUs.

EY-Parthenon · Strategy consulting intern, Geneva Sep 2021 – Jan 2022

- Advised consumer-goods, retail and agribusiness clients on technology-led sustainability strategy.

MIT Auto-ID Lab · Research assistant (supervisor: Dr. Brian Subirana) Feb 2020 – Sep 2021

- **COVID-19 screening from phone-recorded forced coughs** (first author, IEEE OJEMB 2020; 700+ citations): AUC 0.97 (98.5% sensitivity, 94.2% specificity) on the official-test subgroup of a 5,320-subject dataset. A 2024 independent study (Coppock et al., *Nature Machine Intelligence*) found that recruitment bias in crowdsourced audio can inflate such results.
- **Alzheimer's detection from spontaneous speech** (first author, Frontiers in Computer Science 2021): the same architecture reaches 93.8% accuracy on the ADReSS test set from raw audio.
- Set up clinical data collaborations with the Government of Catalonia's Department of Health (Spain) and the Universidad de Guadalajara (Mexico), collecting forced-cough recordings from PCR-tested patients.

Earlier Sep 2017 – Jan 2020

- Data scientist, Nutrino Health (acquired by Medtronic, 2018): improved a food-image classifier with weakly labelled social-media data.
- Lead data scientist, ADLA (student start-up): led the team building a computer-vision recommendation chatbot.
- Lead software engineer, InAGlobe Education: co-built a platform matching engineering students with NGO technical problems.

PUBLICATIONS

1. **J. Laguarta Soler**, S. Cheng, E. Liao, S. Wang. An open 10-meter multi-crop map of India from street-level imagery and vision-language models. *Manuscript in preparation*, 2026.
2. **J. Laguarta Soler**, T. Friedel, S. Wang. Combining deep learning and street view imagery to map smallholder crop types. *Proc. AAAI* 38(20):22202–22212, 2024.
3. **J. Laguarta**, F. Hueto, B. Subirana. COVID-19 artificial intelligence diagnosis using only cough recordings. *IEEE Open Journal of Engineering in Medicine and Biology* 1:275–281, 2020.
4. **J. Laguarta**, B. Subirana. Longitudinal speech biomarkers for automated Alzheimer's detection. *Frontiers in Computer Science* 3:624694, 2021.
5. B. Subirana, F. Hueto, P. Rajasekaran, **J. Laguarta**, et al. Hi Sigma, do I have the coronavirus? Call for a new artificial intelligence approach to support health care professionals dealing with the COVID-19 pandemic. *arXiv:2004.06510*, 2020.

Research featured in BBC News, The Economist and World Economic Forum (2020), and MIT News (2020, 2024).

EDUCATION

MIT · PhD, Electrical Engineering and Computer Science — Sep 2022 – Feb 2027 (expected)

Cornell Tech · MEng, Computer Science · merit scholarship — Sep 2020 – May 2022 (leave of absence, fall 2021)

Imperial College London · BEng, Electronic and Information Engineering — Oct 2016 – May 2019

SKILLS

Machine learning	Multimodal learning, vision-language models, VLM data engines and model-generated supervision, self-supervised pretraining, cross-modal distillation, evaluation science and benchmark reliability, distribution shift, national-scale inference, satellite time series, audio and speech
Tools	Python, PyTorch, TensorFlow, Google Earth Engine, Vertex AI and Google Cloud, distributed multi-GPU training, SLURM, SQL, C/C++
Languages	Catalan, Spanish, English, French